双电机扩展板

MG513XP28_12V ×2，配套CoralC5-Core
三串锂电池2800mAh，目标约一小时

核心板接口

L_PWM
L_DIR
MOTOR_RUN
L_nFAULT
L_ISENSE
L_ENC_A
L_ENC_B
V3V3
V5_SYS
R_PWM
R_DIR
R_nFAULT
R_ISENSE
R_ENC_A
R_ENC_B

文件：03_Core_Interface.kicad_sch

左电机与编码器

L_PWM
L_DIR
MOTOR_RUN
L_nFAULT
L_ISENSE
L_ENC_A
L_ENC_B
VM
V3V3

文件：04_Left_Motor.kicad_sch

右电机与编码器

R_PWM
R_DIR
MOTOR_RUN
R_nFAULT
R_ISENSE
R_ENC_A
R_ENC_B
VM
V3V3

文件：05_Right_Motor.kicad_sch

输入、总开关与降压

V5_SYS
VM

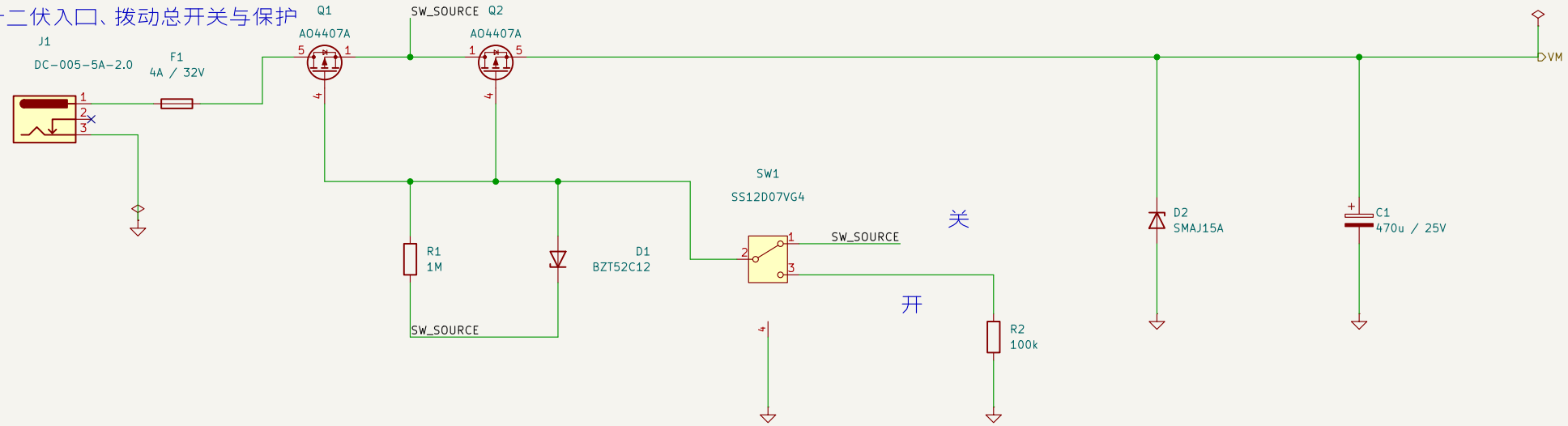
文件：02_Power.kicad_sch

拨动开关切断主电源
两路电机各自限流
编码器用三伏三

原理图设计：实板电流、温升与线束待验证

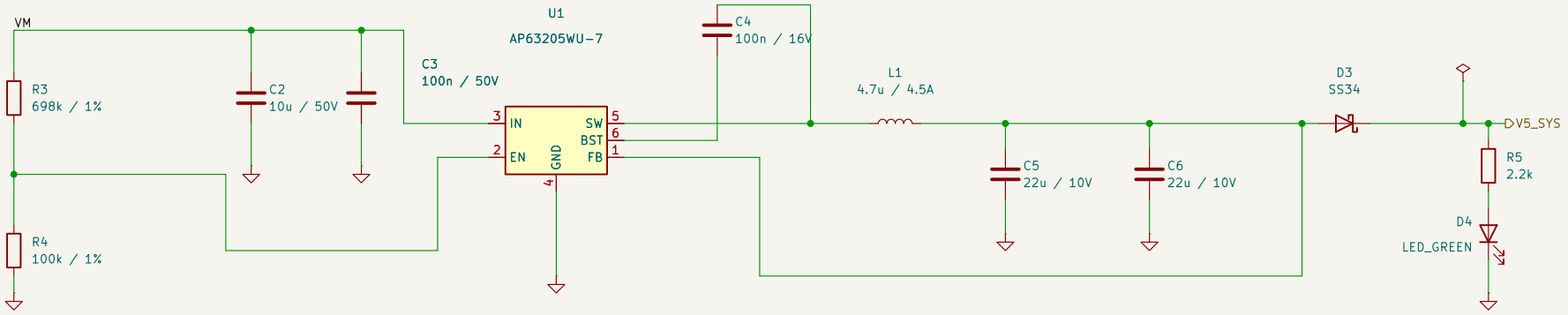
SmallCoral / CoralC5-Motor		
Sheet: /		
File: CoralC5-Motor.kicad_sch		
Title: Coral-C5 双电机扩展板总览		
Size: A4	Date: 2026-10-27	**Rev: M1**
KiCad E.D.A. 10.0.5		Id: 1/5

十二伏入口、拨动总开关与保护



三串锂电池9~12.6V，中心正极；拨动开关控制双MOS主通路
关闭后电机和降压均断电，USB仍可单独供核心板

五伏同步降压与USB供电隔离



五伏负载预算0.75A，核心板入口约4.6~4.9V
按原厂配置4.7uH及两颗22uF，D3隔离USB反向供电

原理图设计：实板电流、温升与线束待验证

SmallCoral / CoralC5-Motor

Sheet: /输入、总开关与降压/
File: 02_Power.kicad_sch

Title: 十二伏入口、拨动总开关与五伏降压

Size: A4	Date: 2026-10-27	**Rev: M1**
KiCad E.D.A. 10.0.5		Id: 2/5

核心板接口

J14A

C5-EDGE-3 / 98P

GND	A1	A1_/_GND	B1_/_GND	B1	GND
AUX_UART_TX	A2	A2_/_PA2	B2_/_PA3	B2	AUX_UART_RX
R_ISENSE	A3	A3_/_PA1	B3_/_PA4	B3	AUX_ADC
L_ISENSE	A4	A4_/_PA0	B4_/_PA5	B4	AUX_GPIO
	A5	A5_/_PH5	B5_/_PA6	B5	
	A6	A6_/_VREF_MCU	B6_/_PA7	B6	
	A7	A7_/_PH4	B7_/_PC4	B7	
	A8	A8_/_PC3	B8_/_PC5	B8	
	A9	A9_/_PC2	B9_/_PB0	B9	
	A10	A10_/_PC1	B10_/_PB1	B10	
	A11	A11_/_PC0	B11_/_PB2	B11	
NRST	A12	A12_/_NRST	B12_/_PE7	B12	
	A13	A13_/_NC	B13_/_PE8	B13	
	A14	A14_/_NC	B14_/_PE9	B14	
	A15	A15_/_PC15	B15_/_PE10	B15	
	A16	A16_/_PC14	B16_/_PE11	B16	
	A17	A17_/_PC13	B17_/_PE12	B17	
	A18	A18_/_PH15	B18_/_PE13	B18	
	A19	A19_/_PE6	B19_/_PE14	B19	
R_nFAULT	A20	A20_/_PE5	B20_/_PE15	B20	
L_nFAULT	A21	A21_/_PE4	B21_/_PB10	B21	
	A22	A22_/_PE3	B22_/_PB12	B22	
MOTOR_EN_REQ	A23	A23_/_PE2	B23_/_PB13	B23	
	A24	A24_/_PE1	B24_/_PB14	B24	
	A25	A25_/_PE0	B25_/_PB15	B25	

J14B

C5-EDGE-3 / 98P

	A26	A26_/_PB9	B26_/_PD8	B26	
	A27	A27_/_PB8	B27_/_PD9	B27	
	A28	A28_/_PH2	B28_/_PD10	B28	L_DIR
AUX_I2C_SDA	A29	A29_/_PB7	B29_/_PD11	B29	R_DIR
AUX_I2C_SCL	A30	A30_/_PB6	B30_/_PD12	B30	L_PWM
	A31	A31_/_PB5	B31_/_PD13	B31	R_PWM
	A32	A32_/_PB4	B32_/_PD14	B32	
L_ENC_B	A33	A33_/_PB3	B33_/_PD15	B33	
	A34	A34_/_PD7	B34_/_PC6	B34	R_ENC_A
	A35	A35_/_PD6	B35_/_PC7	B35	R_ENC_B
	A36	A36_/_PD5	B36_/_PC8	B36	
	A37	A37_/_PD4	B37_/_PC9	B37	
	A38	A38_/_PD3	B38_/_PA8	B38	
	A39	A39_/_PD2	B39_/_PA9	B39	
	A40	A40_/_PD1	B40_/_PA10	B40	
	A41	A41_/_PD0	B41_/_PA11	B41	
	A42	A42_/_PC12	B42_/_PA12	B42	
	A43	A43_/_PC11	B43_/_PA13	B43	SWDIO
	A44	A44_/_PC10	B44_/_PH3	B44	
L_ENC_A	A45	A45_/_PA15	B45_/_NC	B45	
SWCLK	A46	A46_/_PA14	B46_/_NC	B46	
V3V3	A47	A47_/_+3V3	B47_/_+3V3	B47	V3V3
V5_SYS	A48	A48_/_+5V_SYS	B48_/_+5V_SYS	B48	V5_SYS
GND	A49	A49_/_GND	B49_/_GND	B49	GND

调试与扩展

J14C

C5-EDGE-3 / 98P

GND	50	MP1_/_GND	MP2_/_GND	51	GND

J8

SWD / XH2.50

V3V3	1	2	
SWDIO	3	4	
SWCLK			
GND			

J9

AUX

V3V3	1	2	GND
AUX_I2C_SCL	3	4	AUX_I2C_SDA
AUX_UART_TX	5	6	AUX_UART_RX
AUX_ADC	7	8	AUX_GPIO
V5_SYS	9	10	NRST

硬件停机回路

J10

RUN / STOP_NC

R6

100R

MOTOR_EN_REQ

MOTOR_RUN

R7

10k

MOTOR_RUN

V3V3

J10装短路帽，或接常闭停机开关
开路、复位时默认停止电机

严格沿用C5-EDGE-3针序
空闲触点明确标记不连接
该母座不能插计算机PCIe

编码器及三伏三外接负载
合计不得超过100mA

原理图设计：实板电流、温升与线束待验证

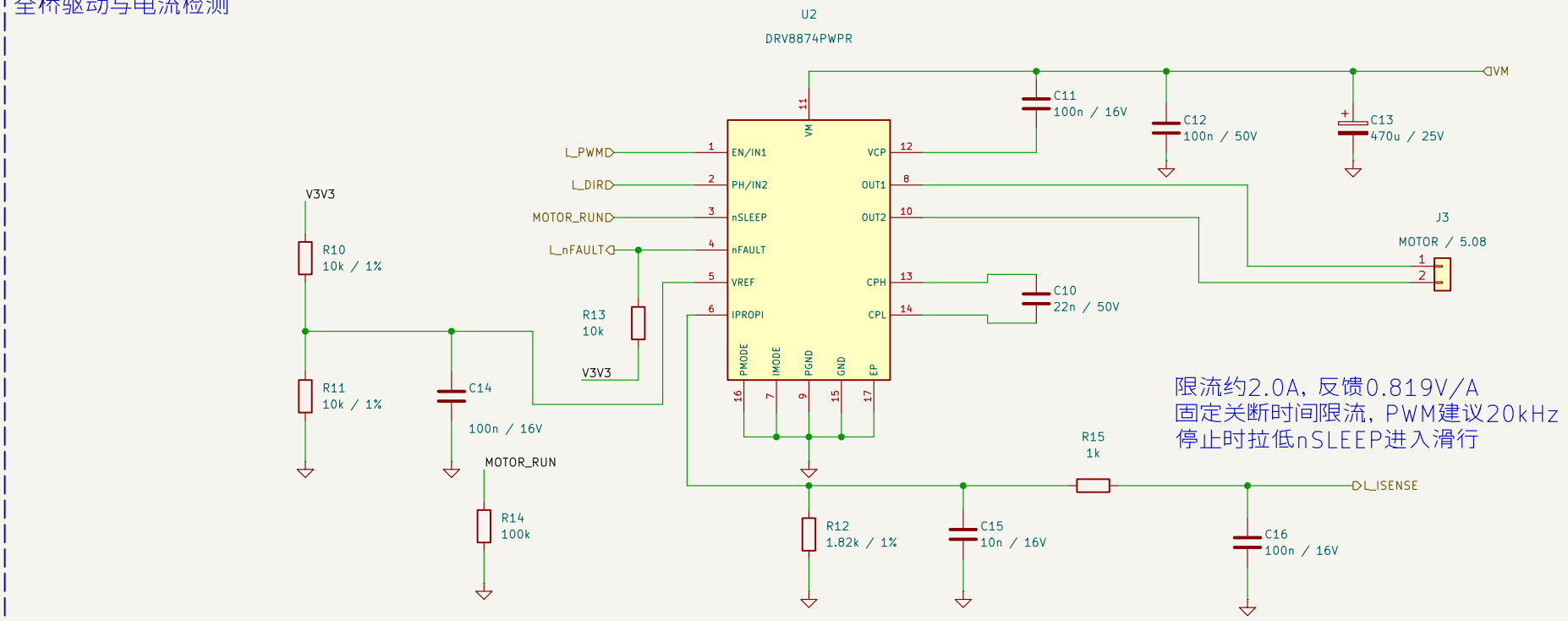
SmallCoral / CoralC5-Motor

Sheet: /核心板接口/
File: 03_Core_Interface.kicad_sch

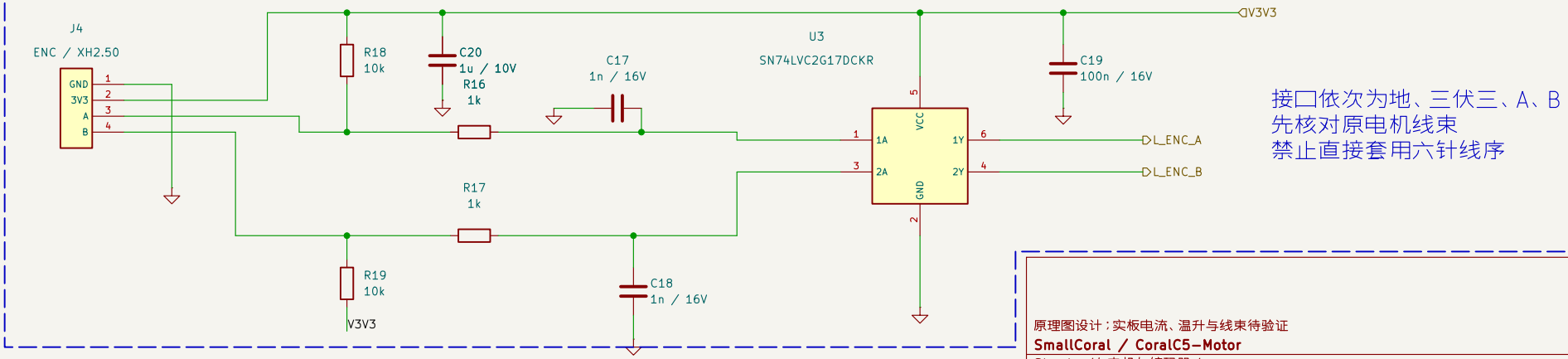
Title: C5-EDGE-3 核心板接口与调试

Size: A4	Date: 2026-10-27	**Rev: M1**
KiCad E.D.A. 10.0.5		Id: 3/5

全桥驱动与电流检测



编码器供电、滤波与整形



原理图设计：实板电流、温升与线束待验证

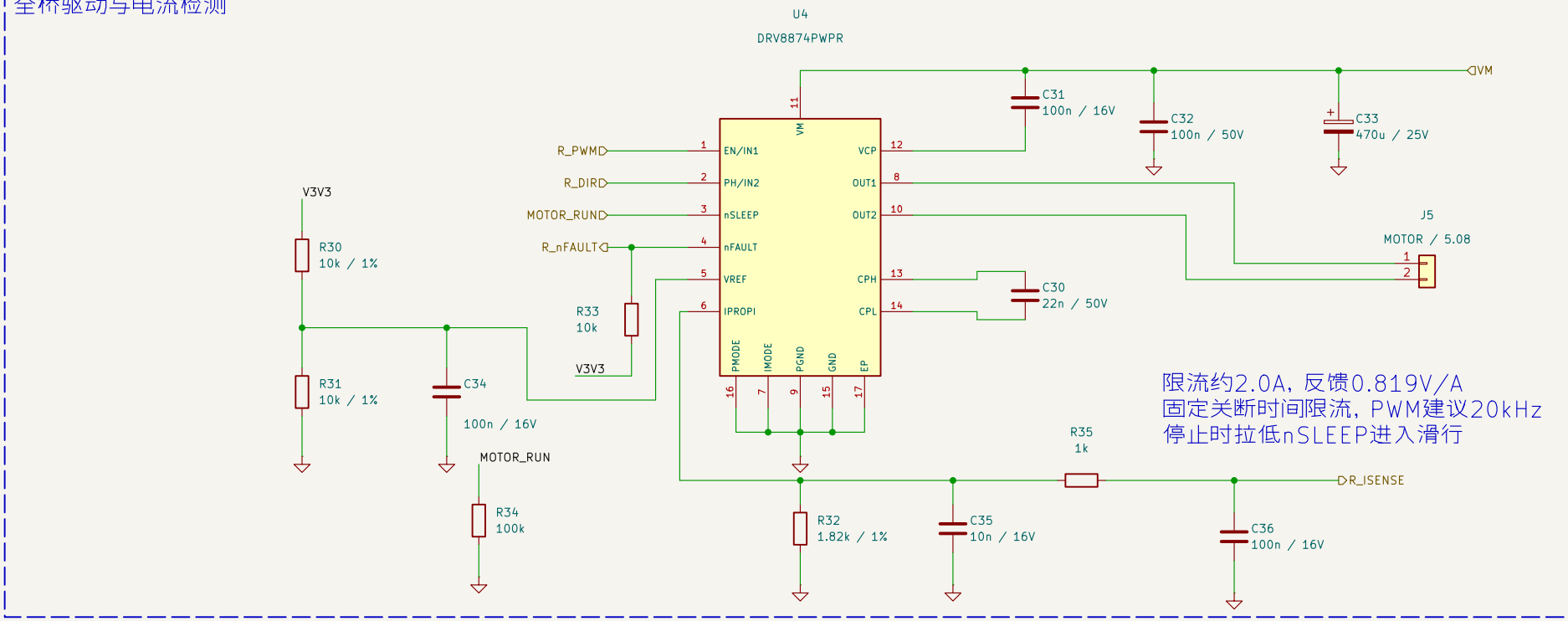
SmallCoral / CoralC5-Motor

Sheet: /左电机与编码器/
File: 04_Left_Motor.kicad_sch

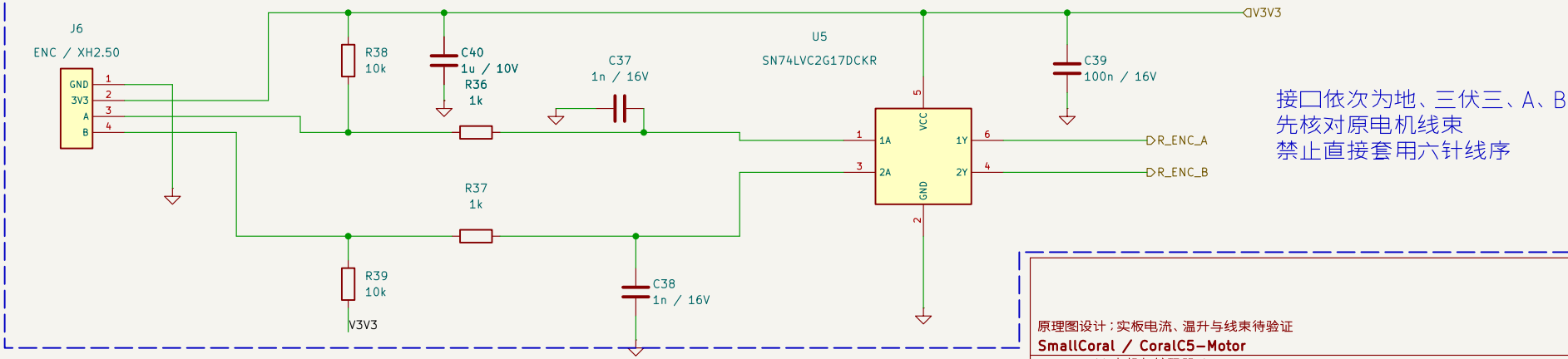
Title: 左电机驱动与霍尔编码器

Size: A4	Date: 2026-10-27	**Rev: M1**
KiCad E.D.A. 10.0.5		Id: 4/5

全桥驱动与电流检测



编码器供电、滤波与整形



原理图设计：实板电流、温升与线束待验证

SmallCoral / CoralC5-Motor

Sheet: /右电机与编码器/
File: 05_Right_Motor.kicad_sch

Title: 右电机驱动与霍尔编码器

Size: A4	Date: 2026-10-27	**Rev: M1**
KiCad E.D.A. 10.0.5		Id: 5/5